

Al-Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Dept.



Microprocessor Systems

3 Credit Hours

13 Dec. 2021

Student Name: [REDACTED]

Mid Exam
30 Point

Question #1: First Question (14 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
C	A	A	B	B	B	A	A	A		A	D	C	A

- Data bus width of Intel 8086 microprocessor is:
A. 16 B. 24 C. 20 D. 32
- One of the following control transfer instruction is not used with signed number.
A. JGE B. JBE C. JL D. JLE
- The segment that contains the assembly language instructions to be executed by the microprocessor is
A. Data segment B. Code segment C. Stack segment D. Extra segment
- Instruction Pointer (IP) contains offset address of _____ segment.
A. Data segment B. Code segment C. Stack segment D. Extra segment
- If the offset address of X in data segment is equal to 105H, and DS=1000H, then the physical address of X =?
A. 10104 H B. 10105 H C. 1105 H D. 11015H
- If offset is a 10-bit number address, then the maximum possible locations are _____.
A. 1024 KB B. 1024 bytes C. 64 KB D. 1 MB
- If the size of memory is 64K, what will be the starting and ending offset addresses of it?
A. 0000H to FFFFH B. 0000H to AFFFH C. 00000H to FFFFFH D. 000H to FFFH
- The content of register BL after execution the instruction MOV BL, -11 :
A. BL=F5H B. BL=FAH C. BL=F7H D. BL=FCH
- In the following instruction sequence, show the value of AL, in binary:

MOV AL, 3FH
NEG AL
XOR AL, FFH

00111111
11000000
11111111
00111110

B. AL= 01011111B

D. AL= 01011111B

10. In the following instruction sequence, show the values of the Carry, and Zero flags :

MOV AL, 00110011b
TEST AL, 8

00110011
00001000
00110011

8 → 1000

cf=0 zf=0

11. Find the content of AL and zero flag (ZF) after execution of the following code:

```
MOV AL, FDH
ADD AL, 3
ADC AL, 3
```

①

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- A. AL=4 ; ZF=0 B. AL=2 ; ZF=1 C. AL=5 ; ZF=0 D. AL=6 ; ZF=1

12. The equivalent code of NOT AX instruction is:

A. ADD AX, 1 B. AND AX, FFFFH C. OR AX, 0000H D. XOR AX, FFFFH

13. If the SP register contains the hexadecimal value F104H, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX, BX
POP AX
POP BX
```

- A. F104H B. 102H C. F104H D. F106H

14. What will be the final value in AX and DX after this code executes?

```
MOV DX, 1
MOV AL, -5
CMP AL, 000H
JB L1
MOV DX, 0
L1: MOV AX, 0
```

0101
1011
....

- A. AX=0; DX=0 B. AX=0; DX=1 C. AX=-5; DX=0 D. AX=1; DX=1

Question #2:

A) Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and DX after execution of the following code?

X DW 35FAH, 48CBH, 231AH, 0080H

MOV BX, OFFSET X

ADD BX, 2

MOV AL, [BX] ; AL = 231AH

INC BX

MOV DX, [BX] ; DX = 0080H

AL = 231AH, DX = 0080H

B) Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and BL after execution of the following code?

X DB 88H, 44H, 50H, 90H, 55H, 23H, 51H

MOV BX, OFFSET X

MOV SI, 0

MOV AL, [BX+4] ; AL = 8CH

MOV CL, X[SI] ; CL = 88H

AL = 8CH

CL = 88H

*C) MOV CL, 03H ; CL = 03H
MOV AL, 127 ; AL = 127D
ROR AL, 01H
JNC L
ADC CL, 00H
L: SHR AL, CL

AX = 00F7H ; CL = 03H ; CF = 1

D) MOV AL, 14 ; AL = E H
MOV AH, 0
MOV BL, 17 ; BL = 11 H
DIV BL

AL = 0 H

AH = E H

10000000
00001000
00011000
8C

(127)
127
00000000
00000000
00000000
00000000

10111111
10111111
11011111
11101111
F7

ZF = 0 PF = 0

AH AL
0E 00

1110
11100001
11H

13 14
12

17 0

Question #3

Suppose we have a microprocessor with 21 bits address bus:

- a) What is the maximum memory capacity will be?

$$2^n = 4$$

- b) For the following instruction `MOV [BX], AL`; what is the addressing mode used?

- Base segment addressing mode

- c) Calculate the physical address of memory location assuming that `DS=8C00h` and `BX=4000h`

$$\begin{aligned} \text{Physical} &= \text{Logical} \times 10\text{H} + \text{offset} \\ 4000\text{h} &= 2000 \times 10\text{H} + \text{offset} \quad \text{offset} = 2 \\ \text{Physical} &= 2000 \times 10\text{H} + 2 = 20002 \end{aligned}$$

- e) Write the equivalent code of `LOOP L` instruction is

$$\begin{aligned} \text{physical} &= \text{logical} \times 10\text{H} + \text{offset} \\ 20000 &= 4000 \times 10\text{H} + \text{offset} \end{aligned}$$

$$\text{offset} = 5$$

$$\begin{aligned} \text{Physical} &= 4000 \times 10\text{H} + 5 \\ &= 50005 \end{aligned}$$